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BioDataHub: An Integrated VS Code Extension for Streamlined Bioinformatics Dataset Analysis and Visualization

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Software

- Review
- Repository
- Archive

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Summary

BioDataHub is an open-source Visual Studio Code extension for discovering, previewing, cataloging, and exploring biological datasets within a familiar development environment. It provides local and online dataset search, dataset download, CSV preview, automatic metadata generation, interactive visualization, principal component analysis (PCA), K-means clustering, data export, and a VCF file opening workflow. By bringing these activities into VS Code, BioDataHub reduces the need to move repeatedly between dataset browsers, standalone viewers, analysis scripts, and visualization tools.

Statement of need

Bioinformatics workflows commonly combine repository search, file inspection, metadata recording, exploratory analysis, and visualization. These activities are often performed with separate web services, spreadsheet applications, and analysis scripts. The resulting context switching makes it harder to preserve the relationship between a dataset, its metadata, and the exploratory results. Although platforms such as Galaxy support reproducible biomedical analysis (Afgan et al., 2018) and Bioconductor provides extensive genomic analysis facilities (Huber et al., 2015), neither is intended to provide a lightweight dataset exploration workflow embedded in a general-purpose code editor.

BioDataHub addresses this gap for researchers and students who already use VS Code or who want a low-configuration entry point for biological data exploration. It is especially useful for inspecting CSV-based gene-expression or other tabular datasets before more specialized statistical analysis. The extension is not intended to replace established domain-specific pipelines, quality-control tools, or statistical packages.

State of the field

Galaxy provides a broad web-based environment for accessible, reproducible, and collaborative biomedical analyses (Afgan et al., 2018). Bioconductor offers a large R-based ecosystem for statistical analysis and visualization of genomic data (Huber et al., 2015). Programmatic visualization libraries such as Matplotlib (Hunter, 2007) and Seaborn (Waskom, 2021) are powerful, but their use generally requires users to write and run separate scripts. Spreadsheet and standalone CSV viewers provide quick inspection but do not normally combine repository search, metadata generation, dimensionality reduction, and clustering in the same interface.

BioDataHub occupies a narrower, complementary position: it integrates dataset discovery and first-pass exploration into VS Code while leaving downstream analysis to the user's preferred

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scientific tools. Its use of the VS Code extension and Webview APIs also makes the workflow available on the same workspace where researchers manage code, data, and analysis notes.

Software design

BioDataHub is implemented in TypeScript and uses the VS Code Extension API, Webview API, and a small set of focused JavaScript libraries. Dataset search, download, CSV preview, metadata generation, visualization, and VCF opening are exposed as VS Code commands and Explorer context-menu actions. The extension also contributes an Activity Bar view for dataset access.

CSV data are parsed for tabular preview and exploratory visualization. The analysis commands include PCA and K-means clustering, using `ml-pca` and `ml-kmeans`, while charts are rendered with Chart.js and D3. Metadata and processed data can be exported for use in later stages of a workflow. The extension supports CSV-oriented exploration and provides a VCF/genotype file opening workflow; it does not attempt to be a complete variant-analysis engine.

Research impact statement

BioDataHub is designed to lower the setup cost of early-stage biological data exploration. Its primary contribution is workflow integration: users can find datasets, inspect tabular content, generate metadata, and produce exploratory plots without leaving the editor in which they organize their analysis. This design can support teaching, exploratory research, and reproducible project setup, while more specialized tools remain available for statistical modeling, sequence processing, and production pipelines.

The repository includes sample biological data and instructions for building the extension locally. Future development can extend format support, repository connectors, visualization controls, and integration with machine-learning workflows.

Quality control

The project is compiled with TypeScript and checked with ESLint. Automated tests are included for extension behavior and PCA-related functionality. The repository's build workflow can be run with `npm install` followed by `npm run build`, and the test workflow with `npm test`.

Availability

The source code is available at https://github.com/mubashir1837/BioDataHub under the MIT License. BioDataHub is also distributed as a Visual Studio Code extension under the publisher `Mubashir-Ali`. The version described here is 1.5.0.

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DRAFT

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